

# Methodology and techniques

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## 1 Aim, terminology and initial concepts

### 1.1 Aim of this document and outcomes

The aim of this document is to provide a framework that will help you to plan and execute a robust, valid T802 investigation. Within this overall aim, the subsidiary aims are to:

- identify the appropriate research methods
- identify the investigative techniques associated with each method
- provide a process for checking the technical validity and practicability of your proposed methodology.

The outcomes of successfully studying the material will be:

- an understanding of the distinction between secondary and primary research
- an ability to make an informed selection of data-collection techniques that will address your research aim and objectives
- a methodology that is technically valid for your research
- a methodology that is practicable and that minimises risk.

The single activity associated with this document is designed to encourage you to apply the various concepts and techniques to your own research. As it is directly relevant to both your research proposal and the work described in TMA 01, I recommend that you do not miss it out.

### 1.2 Terminology

A distinction is made here between the individual *methods* used, such as case study, review and laboratory experiment, and the term *methodology*, which refers to the combination of methods used and how they will be implemented to obtain all the required data. Whilst methods can, to a varying degree, be prescriptive and well defined, the means by which they are used in combination must be a matter for careful consideration. Even if the constituent methods are by themselves valid and useful, a researcher must be sure that, overall, the appropriate range of data will be collected. Methodology therefore concerns what data you need to collect and the sum total of methods you propose to use. Good methodology design will enable you to address your research aim and objectives. The penalty for getting it wrong is that you can put in a lot of effort, then finally realise that certain vital data is lacking.

#### Study note

In the context of T802, it is likely that your methodology will consist of your secondary research plus

one or two primary research methods.

The term 'technique' is used colloquially in a similar manner to 'method'. However, in this context I make a distinction. A method such as a survey could draw upon a variety of techniques. A survey (the method) of six industrial processes might be achieved by six observation visits or alternatively by six telephone interviews (the techniques), depending on the nature of the data sought.

## Study note

This will turn out to be an important point. The way that this document is constructed might imply a sequence of events, in that a researcher decides on the methodology, then the method and finally the technique. This is probably not the best way to look at it now, although it is likely to be the way you present it in your dissertation. In Section 3, I will show how the technique employed follows on from the data needed to meet an objective. This means that the overall methodology should be dictated by the data that is required to meet the objectives.

A survey of stakeholders involved with the locating of a wind farm might use a questionnaire, raising the issue of whether to design a paper version sent by post or an electronic version distributed by email. Similarly, a method such as a case study might entail techniques such as in-depth interviews with functional managers, review of internal documented information and observation of organisational processes. In selecting the technique to employ, a researcher needs to consider a range of practical matters including relevance, cost, speed, reliability and effectiveness.

## 1.3 Primary and secondary research

The various activities that you will undertake during T802 can conveniently be divided into the three categories shown in Figure 1 (also met in Section 3.5 of Preparing for Research). The rectangle represents the totality of work undertaken during the course of T802 research. The area devoted to the three main categories of activity illustrates their changing relative importance over the duration of the research as the work progresses. The three headings shown may be further subdivided into more detailed activities:

- **secondary research:** literature searches, reading, considering
- **primary research:** research design, planning, execution, talking to experts in the field, recording data, data analysis
- **writing:** note-taking, record-keeping, reflecting, summarising, concluding, drafting, drawing, revising, writing up.

It will be clear from this list that there is a degree of overlap between the three sets of activities shown in Figure 1. The relative proportions of the categories should be regarded as illustrative only.

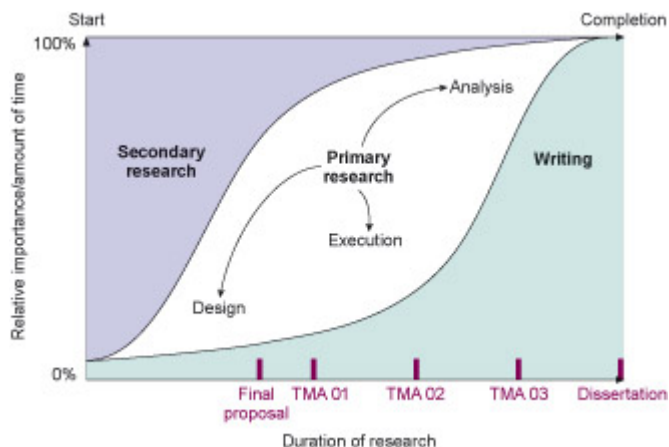


Figure 1 Categories of activity in T802 and their relative importance

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The significance of the scheme illustrated in Figure 1 is that at the heart of T802 (and all research) is the category of activities labelled ‘investigation’. With this in mind, it is worth reminding you of the distinction between the concepts of primary and secondary research.

## Secondary research

This refers to the reading process in Figure 1. It is the search for information through *reading what other researchers have reported*. You are doing this in order to understand the relevant issues and to identify gaps in knowledge, and in due course this will provide much of the information you will write about in Chapter 2 of your dissertation – ‘Research definition’. A significant amount of secondary research must precede primary research because it establishes what the primary research should set out to achieve, as encapsulated in the research aim and objectives.

## Primary research

This refers to your own investigation; the new data collection that you will carry out for the purpose of addressing the research aim you have posed. The design of this research will be described in Chapter 3 of the dissertation – ‘Methodology’. It is important to make a clear distinction between study of the existing work in an area (the secondary research described in the previous paragraph) and the new work involved in the investigation that you will undertake. The relationship between Chapters 2 to 5 of the dissertation template is illustrated in Figure 2.

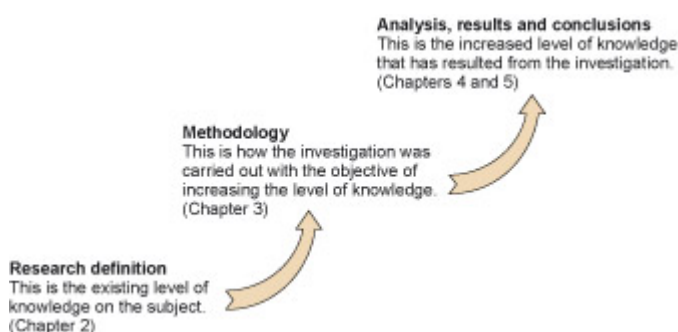


Figure 2 The relationship between Chapters 2 to 5 of the T802 model dissertation

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## 1.4 Quantitative and qualitative research

### Study note

1.4 and 1.5 are covered in more detail in the document Theories and paradigms: their relevance in research.

Research can address many different questions of how, what, when and why. Investigating something that can be observed and measured in a way that can be repeated by other researchers tends to address questions such as 'how often?' or 'how many?' or 'how much?'. This is the long-standing tradition of **quantitative research**, often, but not exclusively, associated with the positivist research paradigm.

However, questions about human behaviour and their social contexts often cannot be quantified, particularly many of the 'why?' questions that explore meaning and that need description and words rather than numbers. The tradition of **qualitative research** applies here, developed by people working in social sciences. Qualitative research is often associated with the naturalistic research paradigm, but, again, not exclusively so.

A lot of research crosses the boundaries of quantitative and qualitative research, using elements of both. As far as T802 is concerned, it is important to note that *both* quantitative and qualitative research traditions have some well-established and rigorous research methods, methodology and data-collection techniques. Mixed methods have also been developed that bring quantitative and qualitative approaches together. Make sure you consult relevant literature to check that you use appropriate methods, methodology and techniques, and that you are using them in an appropriate way.

Further discussion can be found in Yates (2004, pp. 147–53) and Remenyi et al. (1998).

## 1.5 The researcher as stakeholder

It is likely that you will have some involvement with the subject of your research beyond the investigation itself. For example, you may be the manager of a process you are seeking to improve or have a long association with the organisation on which your research is based. This means that you have a stake in the research subject beyond merely the quality of the research that will be carried out. This is both a strength and a weakness, and it is important to recognise both aspects.

Since you already have an interest in, and knowledge of, the research subject and context, you are likely to have insights into the causes of problems and the factors that have an impact on them. You may consider that you implicitly understand, perhaps, how to go about investigating the problem to achieve the best results. This, however, only gives you a potential advantage over a researcher who comes into the field 'cold'. Whilst it is entirely appropriate to take full advantage of your inside knowledge, it is important to recognise the corresponding downside of this situation. Your existing knowledge may lead to a lack of objectivity, a tendency to make subjective judgements instead of evaluation based on sound evidence, or a distortion of the events being investigated as a result of your participation. It is, therefore, important to

recognise how your role beyond that of researcher could affect the research itself, and to take steps to avoid potential pitfalls. For example, you might seek verification of your perceptions from a third party, or gather objective evidence to 'prove' that your own subjective view is correct. Some of the principles of action research (see Section 2.1) may be relevant.

### Study note

Just to emphasise the cautionary note here. What this section is saying is that the possession of heuristic or experiential knowledge is not a substitute for a scientific approach. You will, typically, still need to do as much secondary research for the purpose of situating the research within the body of knowledge. In addition, you will need to validate your approach to the primary research with reference to existing, accepted methods.

## 1.6 Design of primary research

The research design process consists of the four stages shown in Figure 3. Sections 2 and 3 will explore the activities and issues involved in each stage, with the aim of underpinning the development of a robust methodology for your primary research.

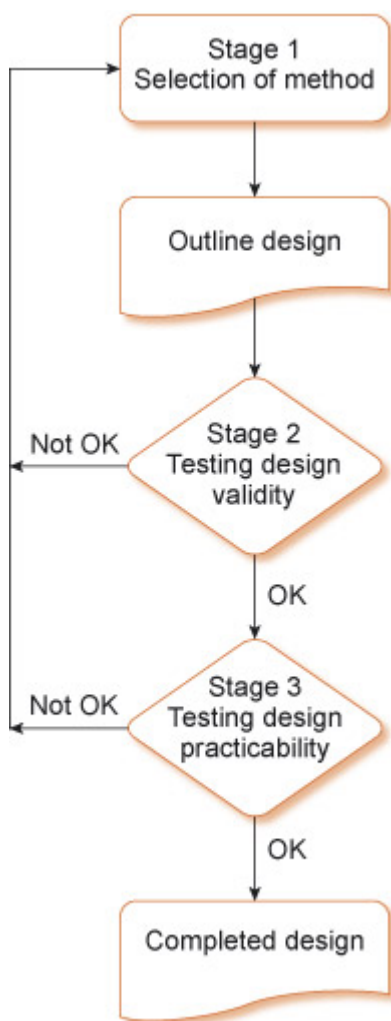


Figure 3 The research methodology design process

You will notice that the design process is presented as a set of steps that are essentially sequential in character. It is to be expected that the day-to-day realities of research will differ from the stages presented in Figure 3; the process is likely to be even more iterative than indicated. However, the benefits of using the model in Figure 3 as a *reference framework* are that it:

- provides the basis for planning
- acts as a common 'language'
- enables progress to be discussed
- acts as a checklist to ensure that all necessary tasks have been undertaken.

Before you embark on the detailed design of your research investigation, it is vital to have a clear definition of the research problem you are intending to tackle. In turn, this requires you to have done sufficient secondary research to know that a gap in the existing knowledge exists. For your research proposal to be accepted you will only have to demonstrate a partially completed design – enough to indicate that it is *likely* that the design will address the research aim. For TMA 01 significant progress will need to be demonstrated and by TMA 02 it should be finalised.

## 1.7 Summary of Section 1

- There is a distinction to be made between methodology and method. The former describes the entirety of your approach to addressing the research aim. The latter refers to a particular way of conducting your research such as a survey. A research technique is a further sub classification of method and should be thought of as the description of the way that data gathering takes place in a given context.
- Primary research is the process of collecting data which is original and turning that data into information and then new knowledge. Secondary research is the search for information by reading what other researchers have reported. Although there will be constant iterations between the two over the course of your research, in the main, secondary research precedes primary research. There should be a clear link between the two with the secondary research informing the primary research.
- Both quantitative and qualitative research methods can be appropriate for T802; possibly even a mixture of the two. The important factor with any methodology is that it is rigorous and fit for purpose.
- You need to be aware of the pitfalls of introducing bias into your research because of your own position as stakeholder.
- The process of research design is iterative. The final design will only become completely clear by TMA 02.

## 2 Research (investigation) methods

It is essential that, from an early stage in the preparation of your research proposal, you consider how you might go about the data gathering and analysis that will be involved in your investigation. This section gives an overview and further sources of information are given in the Additional resources section of the module website. In addition, you should also seek to identify possible methods/techniques in your own literature review by considering what other researchers have done in similar areas.

T802 research can be classified as:

- **Experiments:** practical tests of theory, typically in a laboratory or factory but could also be in 'field' trials.
- **Case studies:** investigations into the occurrence of a phenomenon – which might be a trend, a concept or the use of a technique – within a specific context and using multiple sources of information.
- **Surveys:** systematic collection of information about a representative set of subjects, using a variety of techniques including questionnaires and interviews.
- **Reviews:** either a reassessment of existing work in a subject or re-evaluation of it in the light of new information.

Data gathering that involves practical work of any kind, whether in a laboratory, office or outdoor situation, is referred to as empirical. This applies to the first three methods listed above, but not to review research.

All four methods are expanded upon in the following subsections.

## 2.1 Experiments

Experiments are sometimes based around a hypothesis, in the form of a statement of a relationship to be tested. For example:

- (i) The use of neural networking improves the accuracy of production-cost estimates when compared with existing methods based on linear regression.
- (ii) The use of reagent A is more effective than reagent B in the reduction of *Cryptosporidium* parasite in solution at a specified temperature range.

The research then gathers data that enables the experimenter to evaluate the extent to which the postulated hypothesis is valid. It is almost always possible to express a hypothesis in the form of an open-ended research objective:

- (i) To discover whether the use of neural networking improves the accuracy of production-cost estimates.
- (ii) To decide which of reagents A and B is more effective in the reduction of *Cryptosporidium* parasite, and under what temperature conditions.

Open-ended research objectives are slightly simpler and more direct in expression, so you are encouraged in T802 to use them in preference to posing a hypothesis.

Experiments are also often based around 'What if...?' questions. They share some characteristics with conventional feasibility studies in that they seek an answer to the question 'If we did X, what would result?'. They are clearly focused or deductive in character, in that a specific kind of answer is sought and anticipated. Occasionally, laboratory research has the character of 'let's do this experiment and see what

happens', but you are strongly advised to avoid highly speculative experimentation for T802 purposes, as it can could result in much time being wasted in achieving minimal results.

Experiments can be categorised as either **abstracted** or *in situ*.

- **Abstracted** experiments are undertaken in a laboratory or an analogous setting such as a field trial ground. The subjects of the experiment are taken from their normal setting and subjected to closely controlled experimental procedures. One aim of the process of abstraction is to exclude factors that might influence the results of the experiment and so threaten the validity of the conclusions.
- *In situ* experiments are undertaken in the place where the phenomenon under study occurs. *In situ* experiments are frequently encountered in technological research, though it should be appreciated that setting them up can require a lot of time, effort and expenditure.

If you are considering such an experiment, you must carry out a detailed assessment of the resources required and any inherent risk.

Example 1: testing whether the requirement for segregation of recyclable domestic waste into paper-based, plastic and metal would result in less material being collected when compared with non-segregated collection.

Example 2: testing the extent to which the reorganisation of a particular manufacturing assembly line could improve the quality of the finished product.

- **Action research** is a particular form of practical *in situ* experimentation, in which the researcher is a key participant in the operation being studied. An example might involve changing the way a team operates a manufacturing cell where the researcher is also a member of the team. The issue for a researcher who carries out action research is that it involves one person in two roles: being the experimenter as well as the subject (or one of them) being observed. It can be tricky, but not impossible, to keep the roles distinct. Use of action research also raises another problem – the linear nature of the steps in the process mean that action research can take too long for the limited T802 timescale. Nevertheless if these limitations are taken into account, action research is viable for T802 research.

Further discussion on experiments can be found in Wadsworth (1997), Flowerdew and Martin (2003), Coghlan and Brannick (2001) and Stringer (2007).

A suitable alternative to a real experiment might be to carry out a preliminary study, based on analysis of current 'live' data, followed by detailed planning for a projected situation and a prediction or simulation of how it might perform. Such 'thought experiments' are essentially imagined experiments to test a principle or theory. They can be used as precursors to practical experiments or as investigations in their own right. There are many historically important examples in the fields of science and philosophy. A thought experiment might be a way of exploring the ethics of a particular course of action, for example, or the consequences of making a change to a process.

## 2.2 Case studies



Case study research has distinctive characteristics that have led to a general acceptance that it forms a separate type of research. This view has been put forward most notably by Yin (2003), from which the following definition is adapted (pp. 13–14).

A case study is an empirical [that is, practical] inquiry that

- investigates a contemporary phenomenon within its real-life context, especially when
- the boundaries between phenomenon and context are not clearly evident, [and that]
- relies on multiple sources of evidence.

For further discussion of this, see also Yin (1981a, pp. 97–114; 1981b, pp. 58–65; 2003).

This definition not only helps us to understand case studies but also distinguishes them from other research methods.

Case studies in T802 involve the observation of real, live situations that are relevant to the problem or issue being investigated, and potentially cover a wide scope. They might involve an office, a section within a factory, an organisation, a community or a group of people with a shared aim.

Past T802 studies have featured all of these, plus the more unusual case study environments of an island, an airport, fish in a lake, New Zealand pest infestation, fuel usage in a crematorium, a chemical industry joint venture and oil pollution in a sea location.

There are three categories of case study: exemplar, cross-sectional and longitudinal.

- An **exemplar** case study examines the application of theory in an unusual, critical or interesting setting.
- **Cross-sectional** case studies are used to explore the extent to which a characteristic or phenomenon occurs in a set of people, organisation or other subject at the current point in time.
- **Longitudinal** case studies examine the development of a phenomenon over a period of time.

All three types are encountered in and are relevant to T802-type research. Note, however, that a longitudinal case study might require access to data for a long period, whereas the scope of T802 allows observations to be made only over a comparatively short period. Hence you will need to discuss the possible use of a longitudinal study carefully with your supervisor.

## 2.3 Surveys

According to Thomas, a survey is:

a procedure in which information is collected systematically about a set of cases (such as people, organisations, objects). The cases (or sample units) are selected from a defined population and the aim is to construct a data set from which estimates can be made and conclusions reached about this population

(Thomas, 1996)

Surveys often seem to offer a simple, clear-cut way of gathering primary data on a particular topic. However, behind every successful survey is a lot of hard work, both at the planning and at the execution stage. Do not underestimate the difficulties associated with surveys.

Thomas (1996, pp. 115–24) argues that there are four basic operations involved in undertaking a survey:

- selecting a sample of units from a defined population
- developing and testing standardised ways to measure the units
- applying them to the sample units
- making inferences about the population from which the sample was selected.

Although the term 'survey' is often used colloquially, to mean the same as a questionnaire, it is used here in the wider sense as defined above by Thomas (1996). A questionnaire, whether using paper, internet or telephone, is one technique for carrying out a survey, but is not synonymous with it. Other techniques, such as interviewing and internet searching, might alternatively be used for a survey. In addition there are other techniques which, although not strictly adhering to Thomas's definition, can be regarded as being part of the survey category. These are the Nominal Group, Focus Group and Delphi techniques, which will be covered in more detail in Section 3.1.

A survey always involves the systematic collection of data about a sample of subjects selected from a population, the aim being to identify the characteristic of the population as a whole from the relatively small sample of data collected. The main advantage of surveys is their efficiency in comparison with gathering data from the whole population (as is done in a census). However, there are also significant theoretical and practical problems in selecting the sample and identifying the data sought.

There are three categories of survey: cross-sectional, longitudinal and discovery.

- **Cross-sectional surveys** are used to explore the extent to which a characteristic or phenomenon occurs in a set of people, a single or group of organisations, or other subjects at the time of the survey. They may also be used to test the application of theory to contrasting sets of circumstances. Such surveys are potentially applicable to T802.
- **Longitudinal surveys** involve repeated sampling at different intervals and have the aim of identifying the development of a characteristic or phenomenon over a period of time. They may be contemporary or historical. It would be possible, for example, to undertake a survey of environmental statutes regarding sites of special scientific interest for the period 1994–2004. Whilst not totally inappropriate to T802, longitudinal surveys are unlikely to be of use in primary research because of the short timescale involved.
- **Discovery surveys** are more speculative than the other two. You may not be entirely sure what findings you are looking for, so the questions in these surveys tend to be open ended – hence they are often the precursor to other primary data collection. Be wary of attempting a discovery survey for T802, unless you are convinced that there is indeed something out there to be discovered. See also the section on Grounded Theory in 'Theories and paradigms: their relevance in research'.

## 2.4 Reviews

The final category of research is the review approach, which could be:

- **theoretical** – attempting a synthesis of existing research in the chosen area
- **reinterpretive** – either reviewing an established theory in the light of new data, reinterpreting existing data in the light of a new theory or methodology, or combining two or more data sets.

Periodic reviews of research in a particular topic are necessary because of the way scientific discovery proceeds. Occasional 'paradigm' shifts that establish a new theoretical basis for a subject are interspersed with often lengthy periods in which the scientific community undertakes 'normal' science. The aim of normal science is to solve puzzles, not to pursue major conceptual novelty. In a similar manner, most T802 research is technological and concerned with the solving of problems.

In the T802 context, review research is rare but not unknown. It is unlikely to be aimed at producing a paradigm shift, but could realistically aim to produce a novel solution to a problem using, for example, published experimental data from two or more sources. The task would then be to use the totality of the data to solve the problem or give new insights, where the previous individual experimenters had not succeeded. Alternatively, review research might synthesise a new conceptual model that is applicable across a range of different circumstances from several separate published pieces of research that were each undertaken from a very particular limited perspective.

Review research is not the same as a literature review. They are different activities and serve different purposes. Literature reviewing is secondary research for the purpose of understanding the current 'state of the art'. Review research is primary research that creates new solutions from existing knowledge. As such, review research can be even more demanding than empirical research, requiring both access to relevant existing data and a good grasp of the previous research that generated it. It is by no means an easy option. If you are considering undertaking review research then you will need to discuss with your supervisor, as part of the proposal phase, how your plans differ from secondary research.

## 2.5 Summary of Section 2

- There are four basic subdivisions of methods: experiments, case studies, surveys and reviews. Each has various techniques which could be employed and it is important to apply the most appropriate variation to your research problem.
- Words are used precisely in some of the contexts so beware of misunderstanding if a word has another or less precise meaning in everyday usage.
- Some methods are less suitable for T802 research than others. The following ones should only be used if you are really sure there is not a better alternative:
  - action research
  - thought experiments
  - longitudinal case studies
  - longitudinal surveys
  - reviews.

- Beware of regarding the use of a survey as a 'quick fix' or 'sure fire' way of success. There are significant issues in selecting the sample and identifying the data sought.

### 3 Research methodology design

The first stage of the design problem consists of making informed decisions about:

1. choice of method – empirical (experiment, survey, case study) or review
2. choice of investigative technique(s).

A good statement of the methodology used in a particular piece of research will start with a discussion of the choices made at each level. The options at each level of choice are discussed in the following sections. The conceptual model shown in Figure 4 summarises the complete range of applicable options.

You would be well advised to consider again at this stage what you can bring to your research from your diploma-level studies, especially but not exclusively from the core (compulsory) modules in your programme. You will have met a wealth of theoretical models, problem-solving approaches and analytical techniques, one or more of which could be an important tool for your research. Systems ideas in particular may be very helpful in defining and explaining the particular problem to be addressed. See Additional resources for suggestions and advice relevant to your particular MSc qualification.

#### 3.1 Choice of method and associated techniques

The left hand side of Figure 4 shows the three methods for empirical research and the review approach, subdivided into eleven research *categories*. Taken as a whole, this figure represents a classification of technological research and shows the options that are available. The relationship between methods and the range of possible data-collection techniques is shown on the right hand side of Figure 4.

|                        |                      |            | Technique                                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |
|------------------------|----------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|
|                        |                      |            | Laboratory process<br>Shop floor/administration process<br>Modelling and simulation<br>Analysis of company records<br>Analysis of public records<br>Use of secondary material<br>Questionnaire<br>Focus/Nominal/Depth groups<br>Unstructured interviews<br>Structured interviews<br>Observation and measurement<br>Action research |   |   |   |   |   |   |   |   |   |   |   |
|                        |                      |            | key ○ Often useful △ Sometimes useful □ Occasionally useful                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |   |   |   |   |
| Technological Research | Empirical methods    | Experiment | Abstracted<br>Carried out in a special place                                                                                                                                                                                                                                                                                       | ○ |   |   |   |   |   |   |   |   |   | △ |
|                        |                      |            | 'In situ'<br>Carried out where phenomena normally occur                                                                                                                                                                                                                                                                            |   | ○ |   |   |   |   |   |   |   | △ |   |
|                        |                      |            | 'What if ... ?'<br>Investigates the effects of an alternative state of affairs                                                                                                                                                                                                                                                     |   |   | ○ |   |   |   |   |   |   |   |   |
|                        |                      | Case study | Exemplar<br>Examines the application of theory in a particular setting                                                                                                                                                                                                                                                             |   |   |   | △ |   |   | ○ | △ |   | ○ | □ |
|                        |                      |            | Cross-sectional<br>Investigates differences in a set of phenomena at one point in time in a small number of settings                                                                                                                                                                                                               |   |   |   | △ | △ | △ | ○ |   | △ | ○ | □ |
|                        |                      |            | Longitudinal<br>Investigates differences in a set of phenomena over time in a single or small number of settings                                                                                                                                                                                                                   |   |   | ○ |   | □ | □ | □ | △ | ○ | △ | ○ |
|                        |                      | Survey     | Discovery<br>Uncovers 'what is there'                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   | ○ | △ | △ | △ | □ |
|                        |                      |            | Cross-sectional<br>Investigates differences in a set of phenomena at one point in time in a small number of settings                                                                                                                                                                                                               |   |   |   | △ | △ | △ | ○ |   | △ | △ |   |
|                        |                      |            | Longitudinal<br>Investigates differences in a set of phenomena over time in a single or small number of settings                                                                                                                                                                                                                   |   |   |   | □ | □ | ○ | □ | □ | △ | □ |   |
|                        | Non-empirical method | Review     | Theoretical<br>Synthesises work in a topic, subject or issue                                                                                                                                                                                                                                                                       |   |   |   |   |   | ○ |   |   | □ | □ |   |
|                        |                      |            | Re-Interpretive<br>Reviews theory in the light of new data and information                                                                                                                                                                                                                                                         |   |   |   |   |   | ○ |   |   | □ | □ |   |

Figure 4 The relationship between category and technique

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The techniques under consideration in the T802 context are summarised as follows:

- **Laboratory process** – not common, but appropriate where the problem or issue can be defined in terms of physical or chemical processes. This could also be a prototype build and test, particularly in a computing context, for example.
- **Shop floor/administration process analysis** – often encountered in research related to the control of manufacturing and to supporting administration.
- **Modelling and simulation** – often used, including real and conceptual models. Computer modelling and simulation is an appropriate and useful technique.
- **Analysis of company records** – encountered in research involving management, financial and policy issues. It may have confidentiality implications if the records are not in the public domain.
- **Analysis of public records** – encountered particularly in matters of legislation and in standards documents.

- **Use of secondary material** – this refers to reports by other investigators and is mainly applicable to review research.
- **Questionnaire** – often used. Useful where a range of facts and opinions can be obtained through straightforward questions to a relatively large number of subjects, at least in comparison with the number that can be interviewed.
- **Focus group / Nominal group** – potentially useful group interaction techniques. Both are applicable where discussion amongst experts might produce new ideas or a consensual view on an issue. The nominal group technique has its origins in decision making but it also has value in research. It tends to have more formal procedures than the focus group technique.
- **Delphi** – an asynchronous and anonymous form of the focus group. Therefore likely to be more robust.
- **Unstructured and semi-structured interviews** – useful where an expert can provide useful information through discussion and prompting, rather than just answering set questions.
- **Structured interviews** – useful where the views of a number of experts are sought and can be obtained from a set of questions. Often, structured interviews are used in conjunction with another technique. For example, following a questionnaire, which gives breadth, another technique is used to explore something in more depth.
- **Observation and measurement** – essential where detailed data is required from a wide range of processes, including factory operations and natural and social environments.
- **Action research** – occurs mainly in real-life experimentation; difficult to control and evaluate since the researcher is both an observer and a participant.

You should find Figure 4 useful in narrowing down the potential techniques for your chosen method. However, the applicability of each technique is only one factor in the choice – other factors will be considered during the two testing stages of the research design process, described in Section 3.3. It is also important to consider the choice of data-gathering techniques in relation to the way that you intend to analyse and manipulate the data. Knowing what you are going to do with the data once collected must therefore be an integral part of your decisions about data-collection techniques.

**Example 1:** Pat J's research aim was to forecast the adoption of a particular technology. She decided to make use of the analytical tool **graphical extrapolation**, which required the acquisition of historical data from published sources, and then validated that data for currency using the Delphi method. The latter involved the use of two rounds of questionnaires completed anonymously by appropriate experts.

**Example 2:** The aim of Tarek P's research was to improve an existing industrial process. He decided to make use of **process maps** as an analytical tool, both to describe the current practice and to develop and communicate the design of the improved process. This entailed use of three data-acquisition techniques: observation of people's behaviour in operating the process, interviews with people involved in the process, and physical measurements of materials used through the various stages of the process.

You are likely to have met analytical tools and perhaps also data-collection techniques specific to, or commonly used within, your own subject area during your postgraduate diploma-level studies. It is as

important to consider all the relevant sources of information about research methods applicable to your research as it is to research the subject itself. You should aim to become an expert in your chosen research method/techniques as well as in the subject matter of your research.

### 3.2 Methods, tasks and deliverables

In the final section of ‘Preparing for research’ you identified the objectives for your research. It was emphasised that these objectives should be method independent. You should now be ready to begin to unpack those objectives and to look at how you will achieve them.

The first thing that you will need to do is to identify the data that you will need to collect. Then identify the method, and associated technique, which will deliver the data – I will call the actions that you will need to do to undertake the method **tasks**. Tasks should be described using strong, clear verbs like: *read, interview, plan, perform, specify, implement, design, test, analyse*.

Finally in this sequence you should identify **deliverables**. A deliverable should specify exactly what is to be the result when some work has been completed. Examples of deliverables include:

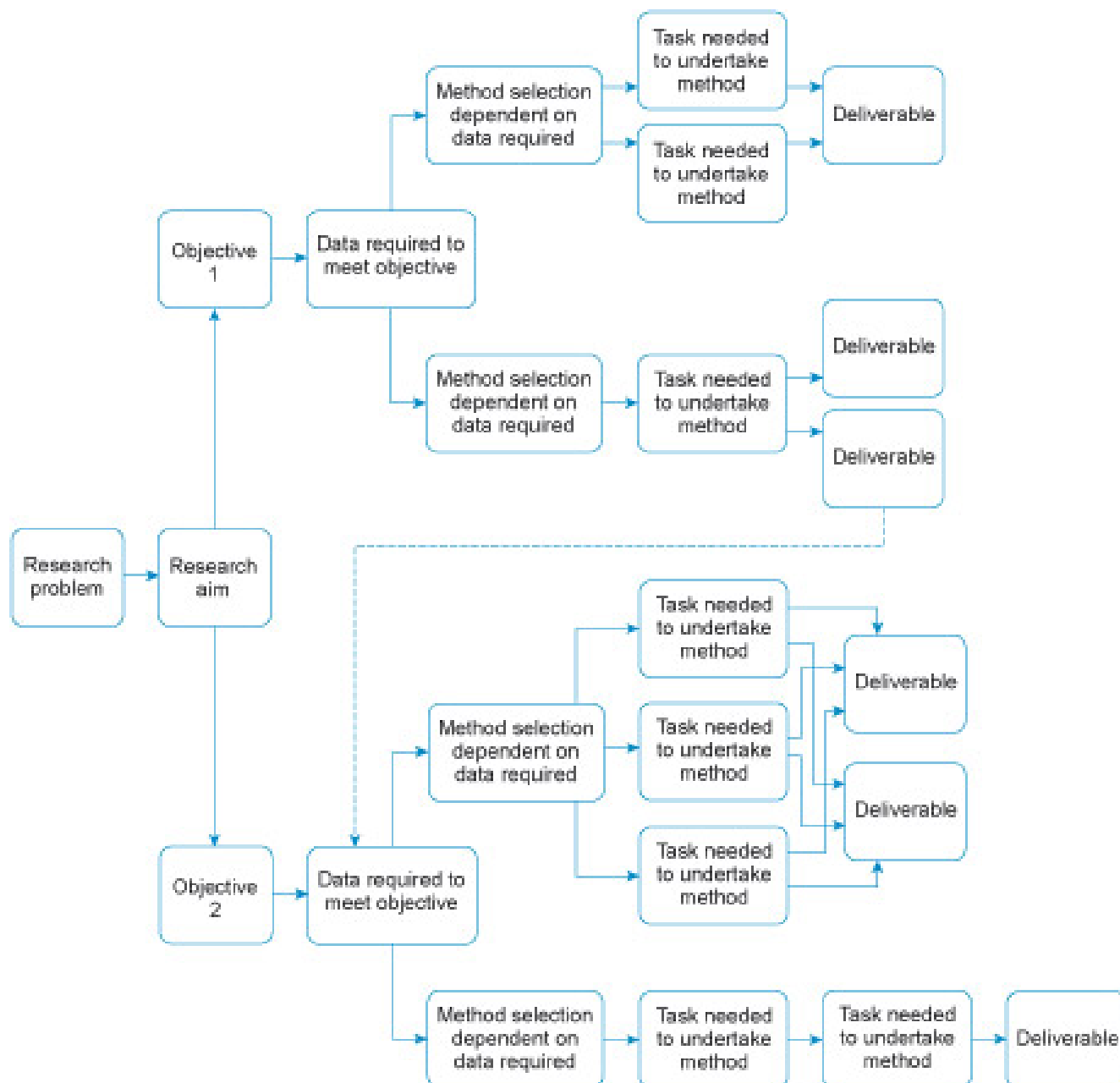
- books and journals obtained and read
- notes taken
- tests performed
- software written
- TMAs written
- a write-up of your literature survey
- a questionnaire developed
- data gathered
- data analysed
- any hardware assembled
- conclusions reached.



Figure 5a

 Maximise

Show description 



**Figures 5a and 5b** The links between research problem and deliverables

[Maximise](#)

[Show description](#) ▼

When you have your list of intended deliverables you then need to ask yourself how achievable *each* of the tasks is for you, given your circumstances and environment. Only take on what you are sure you can do. If, for example, your topic involves robotics but you need to build a robot or write the controlling software from scratch and you've never done this or anything like it before, *don't take this on*. Inevitably, problems will arise that will take a great deal of your time and trouble to sort out – time that you desperately need elsewhere, not in trying to sort out ancillary problems.

So some key questions are:

- What resources do I need to do these tasks? Have I got or can I acquire the hardware, software, access to a laboratory, access to subjects for any interviews or surveys?



- How long will each of these tasks take me?

When you are able to assure yourself (and convince your supervisor) that what you are intending to do for your primary research is feasible within the time constraints of T802, you will be well on your way to an appropriate research proposal. Activity 1, at the end of this document, is an important feature of this proposal. There are, however, just two more sets of issues to consider before it is appropriate to undertake that activity – methodology validity and methodology practicability. These are dealt with in Sections 3.3 and 3.4.

### 3.3 Methodology validity

This subsection represents a 'change of gear' in the sense that the first part looks at some new concepts in significant detail, whilst the second part takes the results of that and turns them into questions. It is OK to skim the information in 'Criteria for methodology validity' in the first instance and only refer to it again if the questions in 'Testing methodology validity' need elucidating in any way.

#### Criteria for methodology validity

The aim of a good methodology is to lead to valid results and conclusions. Ultimately, it must:

- address the research aim
- be executed in accordance with best practice with any variations justified
- produce conclusions that are generalisable; insofar as it is possible to reconcile this with the research paradigm applicable to your situation.

The overall validity of the methodology may be decomposed into four elements:

1. **construct validity** to express the relationship of the data gathered to the research problem
2. **internal validity** for issues concerned with cause and effect and how it is tested
3. **external validity** to demonstrate the generalisability of conclusions in the wider context of the problem or issue investigated
4. **data validity** which is concerned with the need for the data collected to be accurate, sufficient and relevant.

#### 1. Construct validity

A research design must provide the framework and processes that will enable the research aim to be achieved. At this stage it is important to establish the potential of the design by asking five basic questions:

1. What are the deliverables essential to achieving the aim of the research?
2. What information is required to produce the deliverables?
3. What data must be collected to provide the information?
4. Will the proposed methodology be appropriate to obtain the data?
5. How can the data (quantitative and/or qualitative) be analysed and presented effectively?

The construct validity is really a check that you have put together your methodology logically.

## 2. Internal validity

Internal validity is concerned with the way you select and interpret your data. A simple illustration is that the incorrect use of statistical analysis will lead to an untrustworthy result. A robust design needs to avoid weaknesses that could lead to misleading and inconclusive results. The following examples of threats to internal validity, based on Frankfort-Nachmias and Nachmias (1999, pp. 106–13), should be kept in mind.

### ***Selection bias***

Suppose you were investigating the change in attitude and morale resulting from the introduction of a new technique in an office. You might do this by observing a group that had been affected by the change – the **experimental** group – and comparing that with one which hadn't – a **control** group. In comparing the two groups, you would need to consider whether changes observed would be due solely to the new technique introduced or whether the selection process itself might affect the results. It is essential to ensure that the experimental and control groups are neutral in terms of the variable being investigated.

### ***Passage of time***

This refers to those changes that might be expected to occur with the passage of time, irrespective of the experiment being carried out. Consider, for example, an investigation into the effectiveness of training. If the period of training is long, it might be the case that any improvements observed are due to other factors such as a growing familiarity with the work. This problem could also apply to machines used in a piece of research. Suppose you wished to study the effect of different robot 'arms' over a long period. Repeatability of results and the accuracy of the arms are a function not only of their design; wear and tear caused by normal usage of the robot might also influence the results of the study.

### ***Subject group deficiency***

This is concerned with dropout, which prevents the researcher from obtaining complete information. Suppose, for example, in researching the effects of a new method of selecting suppliers for a range of services, 30 are selected at random for a survey but only 10 agree to take part. It needs to be ascertained whether or not the reduced experimental base will be sufficiently representative to provide consistent results. See Additional resources for more on this.

### ***The measuring process***

This is concerned with the degree to which the measurements used in any testing are comparable. If a change occurs in the instrumentation or its calibration, differences in results may be attributable to this cause rather than to the effects of the phenomenon being investigated. This is an important factor in physical experiments but may also be significant, in a less literal sense, in survey work. Suppose your research design involves carrying out a series of structured interviews. You undertake the interviews over a period of time, during which your experience and attitude may change with respect to your investigation. You cannot be sure that this will not influence the way that you interpret answers to your questions.

Wetherall (1968, pp. 152–8) provides the following sources of personal variation in research:

- the limits of human sensation and perception
- misinterpretation

- effects of repetition, such as practice and fatigue
- irrelevant stimulation (distraction)
- selective, preconceived observation
- memory and time lapses in recording
- motivation.

The greater the degree of reliance on the human recording of research data, the greater the potential for psychological, social and physiological factors to affect the accuracy of the data.

### **Testing (Hawthorne effect)**

This is concerned with the way that the process of testing can itself alter the results of the investigation. For example, if you were testing the change in attitudes to quality created by awareness training, the very act of interviewing or otherwise investigating the members of the control group might change their attitudes. The 'Hawthorne effect', named after the series of experiments led by Elton Mayo at the Hawthorne Plant of the Western Electric Company in Cicero, Illinois, is an example of the problem of testing. Researchers found that the activities involved in the research altered the results obtained. For example, they undertook an experiment to determine the relationship between the level of illumination on the shop floor and production output. They discovered that if the lighting was improved, performance increased as might be expected. Counter-intuitively, they also observed that when the lighting was returned to its previous level of illumination, performance still increased. It was the experiment itself that was producing the change in the dependent variable, rather than the action of the independent variable.

### **3. External validity**

After you have studied Theories and paradigms: their relevance in research you will realise that some research paradigms recognise that some research will be context specific and not amenable to generalisation. However, even if you are faced with this situation you should consider, and state if appropriate, in what ways your research may be of relevance to others.

One of the key aspects of much of the research which follows the positivist paradigm is that it should be possible to generalise the results from the specific setting of the investigation to the subject area as a whole. There are two reasons, however, why such an investigation may not be generalisable in terms of the field of study or class of problems in which it is located:

1. The research problem is ill-defined or does not derive from the question 'What is it important to know?'. This question is, in turn, derived from an analysis of the gap in knowledge. It is important to locate the research problem within the body of existing knowledge. The challenge is then to establish the relevance of the work in a wider context and to reflect that relevance in the research problem.
2. The selection of an atypical unit to investigate. This may mean that the results cannot be generalised to a wider class of problem.

### **4. Data validity**

Data is the raw material from which you will obtain the information required to produce your deliverables. Much useful information will come from observation of behaviour and from verbal comments, neither of which can be quantitatively represented. In planning your data-collection method and considering its validity, you need to ask yourself the following four broad questions:

1. How do I justify the method of collection selected as appropriate for the data required, in comparison with other methods?
2. In what ways will the data collected be relevant, sufficient and accurate for the purpose required?
3. Do I need to consider undertaking a pilot study?
4. Will the sample I have selected be representative of the whole population?

If you can answer these questions convincingly, then you have completed an important task.

## Testing methodology validity

### Questions to be considered

1. Does the proposed methodology enable each and every objective to be achieved? It is useful to list the data needed to address each objective and to identify exactly how the intended methodology and techniques will provide that data. Sub-questions here are:
  - What are the deliverables essential to achieving the aim of the research?
  - What information is required to produce the deliverables?
  - What data must be collected to provide the information?
  - Will the data collected be relevant, sufficient and accurate for the purpose required?
  - Will the proposed methodology be appropriate to obtain the data?
  - How can the data (quantitative and/or qualitative) be analysed and presented effectively?
  - Do I need to consider undertaking a pilot study?
2. Is it appropriate to adopt an experimental/control group approach and establish a comparison between the subject of interest and a control subject?
3. How will selection bias be avoided?
4. Will the 'passage of time' problem be a threat to validity? If so, how can I ensure that it is avoided?
5. Is there a danger that my subject group will be insufficiently representative?
6. Do I need to be aware of side effects from the measuring process?
7. Will it be possible to ensure that testing, interviewing and measuring do not themselves alter the results of the investigation?
8. How can I ensure that, if appropriate for the research paradigm, the results of the investigation will be generalisable? Can what I learn from, say, a single case study be considered applicable to other situations? If not, why not, and what can I do about it?

## 3.4 Testing the practicability of your methodology

So far Sections 2 and 3 have been concerned mainly with 'What is to be done?'. This section concentrates on 'How will it be achieved?'. Almost certainly, while you were reading about the selection of data-collection techniques and assessment of their validity, you also had in mind questions regarding the practicability of following your own methodology. Will it take too long? Will the subjects selected be

prepared to provide the data required at the time you need it? Can you fit in activities such as site visits and interviewing with your other commitments?

Hence, the next stage of the research design process is concerned with how to check that the investigation can be carried out in the time available. This involves detailed planning and evaluation using the Gantt chart that you started in Preparing for research.

There is, however, one final concept that is important before you actually begin work on your own plans; that of **milestones**. A milestone is a major point in your project work that you must pass in order to get to your goal. A milestone is that date by which a deliverable needs to be delivered!

The University sets some milestones such as the dates your TMAs are due, and the date your dissertation has to be submitted. Whilst these milestones are obvious ones to include on your Gantt chart, the ones that you create yourself are just as important.

To do this you will need to use your lists of tasks and deliverables to determine what needs to be done in what order. It is very likely that the start of one of your tasks will be dependent on a deliverable. For example you will have to identify, obtain and read the relevant materials *before* you can write up your literature survey, or you may have to assemble the necessary hardware components *before* you can begin an experiment. Key aspects you will need to consider are:

- What is the date that each deliverable is needed?
- What is the duration of each task?
- What is the relationship between tasks?

An example of milestones and associated deliverables, tasks and methods (similar to Figure 5(b)) is given in Additional resources.

### 3.5 Summary of Section 3

Section 3 is one of the most important sections of the whole of the T802 module materials and it introduces key concepts that you will need both for your research proposal and as your research proceeds. You will note that it is supported by several other parts of the module materials; most notably (but not exclusively) by Theories and Paradigms: their relevance in research and also the material in the Generic methods section and the Additional reading list in the Additional resources.

Key points to note:

- Your choice of method(s) is crucial to the success of your research. This choice should be driven by the data needed to achieve each objective. You may need a combination of methods to form an overall methodology.
- Once you have identified an overall methodology, appropriate techniques should be identified for each associated method. Then, appropriate tasks and deliverables identified. You might want to sketch out a diagram similar to that in Figure 5(b) in order represent how your ideas are developing.

- There are then a series of questions to be considered in order to test the overall validity of the methodology. These questions are not applicable in every case and there may be additional questions which need answering. This is something to discuss with your supervisor.
- Finally there is the concept of a milestone to be considered. This is important in testing the practicability of your methodology. A milestone is a major point in your project work that you must pass in order to get to your goal. Each milestone will have an associated deliverable.

## 4 Developing an outline methodology

Having read Sections 1–3 of this part of T802 and, assuming that you have also completed the activity at the end of 'Preparing for research', you should be in good shape to complete a major aspect of your initial research proposal.

### Activity 1

Activity 1 gives you the opportunity to sum up what you've learned in Section 3. You may find that you need to reread the section as you undertake this activity.

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